

PIYUSH ARORA

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Summary

Electronics and Computer Engineering student with strong foundations in algorithms, ML, and embedded systems. Hands-on experience in EV route optimization, EEG signal classification, and plant health monitoring. Skilled at interpreting APIs and backend logic to extend functionality and deliver practical, research-driven solutions.

Education

B.Tech, Electronics and Computer Engineering, VIT Chennai

Sept 2022 – Sept 2026

CGPA: 8.83

Core Courses: Operating Systems, Data Structures, Algorithms, Databases, Embedded Systems, Analog Circuits, Computer Networks

Skills Summary

Languages: Java, Python, C, SQL, Embedded C/C++

Frameworks: TensorFlow, Scikit-learn, Pandas, Keras

Tools: Git/GitHub, Keil IDE, Folium, ORS/Overpass API, Selenium

Soft Skills: Problem Solving, Teamwork, Communication, Leadership

Internship Experience

EV Incubation, Testing & Research Centre, VIT Chennai

Research Intern

Jul 2025 – Aug 2025

- Analyzed **10–15** pathfinding algorithms and designed a toll-aware A* variant.
- Drafted a comparative report explaining performance differences and API integration details.
- Learned how toll/road-cost data is structured in routing backends and applied this knowledge to cost-effective routing.

Tech Intern

Apr 2025 – Jun 2025

- Implemented Dijkstra, A* and Genetic Algorithms for EV route optimization; visualized and validated results with Folium maps.
- Benchmarked GA efficiency (**3s** runtime on Intel i5) by tuning mutation rates and generations.
- Investigated ORS API documentation to uncover EV-specific parameters and applied them to improve route accuracy.

Projects and Research

Google Translator Automation in Python, Python, Selenium

Jul 2025 – Aug 2025

- Lets the user choose a source language, target language, and a phrase and opens Google translate in browser.
- Extracts the translated text from the webpage using JavaScript executed via Selenium.
- Using raw DOM classes for scraping introduces fragility since these class names are often auto-generated and can change without notice, leading to failures in automation, which is why CSS style semantic attributes are used.

EEG Classification for Schizophrenia, Python, TensorFlow, Keras

Mar 2025 – Apr 2025

- Designed compact **1D CNN** with bandpass filtering and overlapping-epoch segmentation for signal clarity.
- Achieved **70% accuracy** on EEG datasets; examined differences across Alpha, Beta, and Gamma brain waves.

AI-based Plant Health Monitoring (Research Article), Python, Scikit-learn

Jan 2024 – Feb 2025

- Built ML pipeline for plant stress detection; achieved **98% accuracy (KNN)** and **93% F1 (RF)**.
- Extended with RBM + Logistic Regression (**84% acc**); presented work at IIT Roorkee.